1 (text)

$$T(n) = 5T\left(\frac{n}{7}\right) + 3n^2$$

$$= 5\left(5\,T\left(\frac{n}{49}\right) + \frac{3n^2}{7}\right) + 3n^2$$

$$= 25\,T\left(\frac{n}{49}\right) + \frac{3n^2}{7} + 3n^2$$

$$= 25\left(5\,T\left(\frac{n}{343}\right) + \frac{3n^2}{49} + \frac{3n^2}{7}\right) + 3n^2$$

$$= 125\,T\left(\frac{n}{343}\right) + \frac{3n^2}{49} + \frac{3n^2}{7} + 3n^2$$

$$\vdots$$

$$= 5^i\,T\left(\frac{n}{7^i}\right) + \dots 3n^2$$

$$= 5^{\log_7(n)}\,T\left(\frac{n}{\log_7(n)}\right) + 3n^2$$

$$\boxed{\begin{array}{l} T(n) = 5^{\log_7(n)}\,T(1) + 3n^2. \\ T(n) = n^2, \text{ since } n^2 \text{ is greatest power.} \end{array}}$$

o master theorem

$$T(n) = 5T\left(\frac{n}{7}\right) + 3n^2$$

Let $a = 5$, $b = 7$, $d = 2$,

note $5 < 7^2$, therefore we proceed by case 1 hence,

$$\boxed{\Theta(n^2).}$$

2 (text) master Theorem:

a. $T(n) = 5T(\frac{n}{4}) + n^7$

In this recurrence $a = 5$, $b = 4$, $d = 7$. now apply master theorem, by first comparing a and b^d,

$5 < 4^7$

hence we follow case 1: $T(n) = \Theta(n^d)$ if $a < b^d$.

Therefore, $T(n) = \Theta(n^7)$.

b. $T(n) = 3T(\frac{n}{3}) + 7n$

$a = 3$, $b = 3$, $d = 1$.

Apply master theorem,

$3 = 3^1$, so follow case 2: $T(n) = \Theta(n^d \log n)$ if $a = b^d$

Therefore, $T(n) = \Theta(n \log n)$.

c. $T(n) = 15T(\frac{n}{3}) + 100$,

$a = 15$, $b = 3$, $d = 0$ (since constants are expressed as n^0).

$15 > 3$, so we go by case 3: $T(n) = \Theta(n^{\log_b(a)})$.

now consider,

$\Theta(n^{\log_3(15)})$.

Since $\log_3(15) \approx 2.4$

$T(n) = \Theta(n^{2.4})$.

d. $T(n) = 9T(\frac{n}{3}) + n^2$

$a = 9, \quad b = 3, \quad d = 2.$

Since $9 = 3^2$, we apply case 2.

Therefore,

$$T(n) = \Theta(n^2 \log n).$$

e. $T(n) = 6T(\frac{n}{5}) + n^3$

$a = 6, \quad b = 5, \quad d = 3.$

Since $6 < 5^3$, we go with case 1.

Therefore,

$$T(n) = \Theta(n^3).$$

3(Text) ~~BOY,RIS,GUA,ATL,SUC,CES,QUI,PUT,ANT,VAU,HUI,VAL,SAN,SAN,CAQ,VIC,CAL,LA_GUA,~~
~~TOL,META,CAU,BOL,AMA,MAG,CHO,NSA,GUA,CAS,COR~~

0	B	O	Y			
1	R	I	S			
2	G	U	A			
3	A	T	L			
4	S	U	C			
5	C	E	S			
6	Q	U	I			
7	P	U	T			
8	A	N	T			
9	V	A	U			
10	H	U	I			
11	V	A	L			
12	S	A	N			
13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
17	L	A	_	G	U	A
18	T	O	L			
19	M	E	T	A		
20	C	A	U			
21	B	O	L			
22	A	M	A			
23	M	A	G			
24	C	H	O			
25	N	S	A			
26	G	U	A			
27	C	A	S			
28	C	O	R			

⇒

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2	G	U	A			
3	A	T	L			
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5	C	E	S			
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14	C	A	Q			
15	V	I	C			
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23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

⇒

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13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
17	T	O	L			
18	M	E	T	A		
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23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

⇒

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8	A	N	T			
9	V	A	U			
10	H	U	I			
11	V	A	L			
12	S	A	N			
13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
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23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

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22	M	A	G			
23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

⇒

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8	A	N	T			
9	V	A	U			
10	H	U	I			
11	V	A	L			
12	S	A	N			
13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
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23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

0	B	O	Y			
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2	G	U	A			
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4	S	U	C			
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7	P	U	T			
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14	C	A	Q			
15	V	I	C			
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22	M	A	G			
23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

⇒

0	B	O	Y			
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2	G	U	A			
3	A	T	L			
4	S	U	C			
5	C	E	S			
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7	P	U	T			
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13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
17	T	O	L			
18	M	E	T	A		
19	C	A	U			
20	B	O	L			
21	A	M	A			
22	M	A	G			
23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

0	B	O	Y			
1	R	I	S			
2	G	U	A			
3	A	T	L			
4	S	U	C			
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7	P	U	T			
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9	V	A	U			
10	H	U	I			
11	V	A	L			
12	S	A	N			
13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
17	T	O	L			
18	M	E	T	A		
19	C	A	U			
20	B	O	L			
21	A	M	A			
22	M	A	G			
23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

0	B	O	Y			
1	R	I	S			
2	G	U	A			
3	A	T	L			
4	S	U	C			
5	C	E	S			
6	Q	U	I			
7	P	U	T			
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10	H	U	I			
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12	S	A	N			
13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
17	T	O	L			
18	M	E	T	A		
19	C	A	U			
20	B	O	L			
21	A	M	A			
22	M	A	G			
23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

0	B	O	Y			
1	R	I	S			
2	G	U	A			
3	A	T	L			
4	S	U	C			
5	C	E	S			
6	Q	U	I			
7	P	U	T			
8	A	N	T			
9	V	A	U			
10	H	U	I			
11	V	A	L			
12	S	A	N			
13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
17	T	O	L			
18	M	E	T	A		
19	C	A	U			
20	B	O	L			
21	A	M	A			
22	M	A	G			
23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

0	B	O	Y			
1	R	I	S			
2	G	U	A			
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13	S	A	N			
14	C	A	Q			
15	V	I	C			
16	C	A	L			
17	T	O	L			
18	M	E	T	A		
19	C	A	U			
20	B	O	L			
21	A	M	A			
22	M	A	G			
23	C	H	O			
24	N	S	A			
25	G	U	A			
26	C	A	S			
27	C	O	R			
28	L	A	_	G	U	A

L	A _ G U A
M	A G
V	A L
C	A L
S	A N
J	A N
C	A Q
C	A J
V	A U
C	A U
C	E S
M	C + G
C	H O
V	I C
C	I S
A	M A
A	N T
T	O L
S	O L
C	O R
B	O Y
N	S A
A	T L
G	U A
G	U A
J	U C
G	U I
W	U I
P	U T

final array

0	AMA
1	ANT
2	ATL
3	BOL
4	BOY
5	CAL
6	CAQ
7	CAS
8	CAU
9	CES
10	CHO
11	COR
12	GUA
13	GUA
14	HUI
15	LA_GUA
16	MAG
17	me+G
18	NSA
19	PUT
20	QUI
21	RIS
22	SAN
23	SAN
24	SUC
25	TOL
26	VAL
27	VAU
28	VIC

4 (text)

Add the following items [62, 62, 69, 70, 4, 103, 85, 27, 86, 40, 96, 77, 23] to the HashTable in order.

```
int h1 (int key) {
    int x = (|key – 9|) * (key + 13);
    x = x / 15;
    x = x + key;
    x = x % M;
    return x;
```

1) 62 → x = 53 * 75 = 3975

x = 3975/15 = 265

x = 265 + 62 = 327

x = 372 % 7 = 5

```
int h2( v )
    v = v^3
    v = v + 1
    return v
```

					62	
0	1	2	3	4	5	6

2) 62 → collision

$(h1(key) + i * h2(key)) \% M$

h1(key) = home slot
h2(key) = step size
i = collision num

stepsize → $62^3 + 1 = 238,329$ % 7 = 0 ⇒ Rehash

2 * 7 = 14 → 17 next prime

96	86		85		62	69	27	77	4	23	62		103	40		70
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

3) h2(62) → 327 % 17 = 5

collision
h2(62) = 238,329 % 17 = 6

(5 + 1 * 6) % 17 = 11

5) h2(69) → 60 * 82 = 4920

4920 / 15 = 328

328 + 69 = 397
397 % 17 = 6

7) h1(70) → 61 * 83 = 5063

5063 / 15 = 337

337 + 70 = 407
407 % 17 = 16

8) h1(4) → -5 × 17 = -85

-85 / 15 = 5

5 + 4 = 9

9 % 17 = 9

9) h1(103) → 94 * 116 = 10,904

10,904 / 15 = 726

726 + 103 = 829

829 % 17 = 13

10) h1(85) → 76 × 98 = 7448

7448 / 15 = 496

496 + 85 = 581

581 % 17 = 3

11) h1(27) → 18 × 40 = 720

720 / 15 = 48

48 + 27 = 75

75 % 17 = 7

12) h1(86) → 77 × 99 = 7623

7623 / 15 = 508

508 + 86 = 594

594 % 17 = 16 ⇒ collision

h2(86)

(16 + 1 * 636057) % 17 = 1

13) h1(40) → 31 × 53 = 1643

1643 / 15 = 109

109 + 40 = 149

149 % 17 = 13 ⇒ collision

h2(40)

(13 + 1 × 64002) % 17 = 9 ⇒ collision

$(13 + 2 \times 64001) \% 17 = 5 \Rightarrow$ collision

$(13 + 3 \times 64001) \% 17 = 7 \rightarrow$ collision

$(13 + 4 \times 64001) \% 17 = 14$

h1(96) $87 \times 109 = 9483$

$9483 / 15 = 632$

$632 + 96 = 728$

$728 \% 17 = \underline{14} \Rightarrow$ collision

h2(96)

$(14 + 1 \times 884,737) \% 17 = 3$

$(14 + 2 \times 884737) \% 17 = 4$

$(14 + 3 \times 884737) \% 17 = 0$

h1(77) $\hookrightarrow$ $68 \times 90 = 6120$

$6120 / 15 = 408$

$408 + 77 = 485$

$485 \% 17 = 4 \rightarrow$ collision

$(4 + 1 \wedge 456534) \% 17 = 8$ ✓

h1(23) $\rightarrow$ $14 \times 36 = 504$

$504 / 15 = 33$

$33 + 23 = 56$

$56 \% 17 = 5$

h2(23) $\Rightarrow 12,168$

$(5 + 1 \times 12168) \% 17 = 1$ collision

$(5 + 2 \times 12168) \% 17 = 14$ collision

$(5 + 3 \times 12168) \% 17 = 10$ ✓ !!

Final hash table

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
96	86		85		62	64	27	77	4	23	62		103	40		70